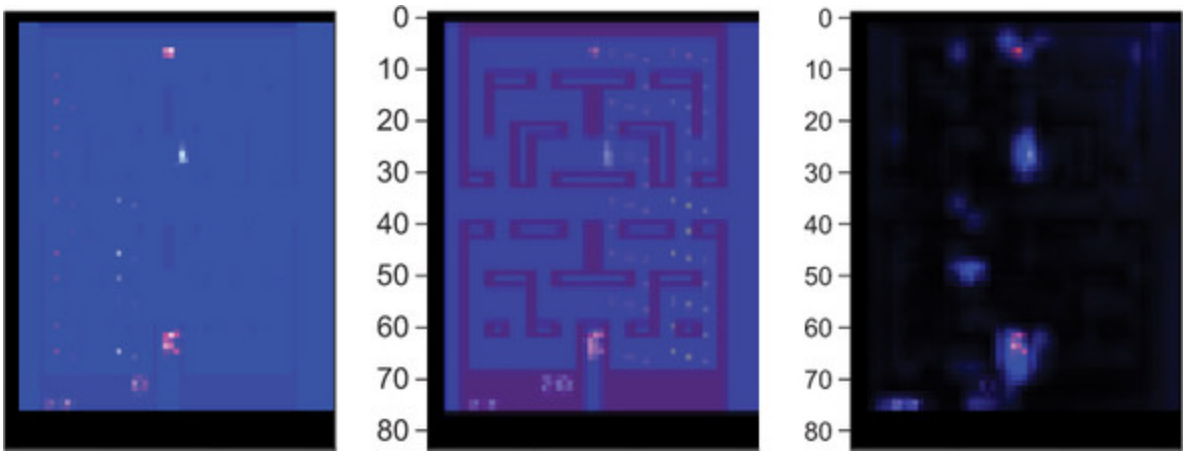


Relational modules using the self-attention mechanism are powerful tools in the machine learning toolkit, and they can be very useful for training RL agents when we want some idea of how they're making decisions. Self-attention is one mechanism for performing message passing on a graph, as we discussed, and it's part of the broader field of graph neural networks, which we encourage you to explore further. There are many implementations of graph neural networks (GNNs) but particularly relevant for us after this chapter is the *graph attention network*, which uses the same self-attention mechanism we just implemented but with the added ability to operate on more general graph-structured data.

Summary

- Graph neural networks are machine learning models that operate on graph-structured data. Graphs are data structures composed of a set of objects (called nodes) and relations between objects (called edges). A natural graph type is a social network in which the nodes are individuals and the edges between nodes represent friendships.
- An adjacency matrix is a matrix with dimensions A:$N \times N$ where N is the number of nodes in a graph that encodes the connectivity between each pair of nodes.
- Message passing is an algorithm for computing updates to node features by iteratively aggregating information from a node's neighbors.